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EXP6:5G NR Frame Structure Analysis with Different Subcarrier Spacing

Aim

To analyze the 5G NR frame structure by studying the effect of different subcarrier spacings,

slot durations, and OFDM symbols using MATLAB.

Objectives

1. To analyze different Subcarrier Spacing (SCS) values used in 5G NR numerology.

2. To compare the slot duration corresponding to different Subcarrier Spacing values in 5G

NR.

3. To study the number of OFDM symbols present in a 5G NR slot for normal cyclic prefix.

Objective1:

Procedure

- Define the supported 5G NR subcarrier spacing values.
- Display the values using MATLAB.
- Plot the subcarrier spacing for different numerologies.

Program:

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

mu = 0:4; % Numerology Index

bar(scs)

grid on

set(gca,'XTick',1:5)
```

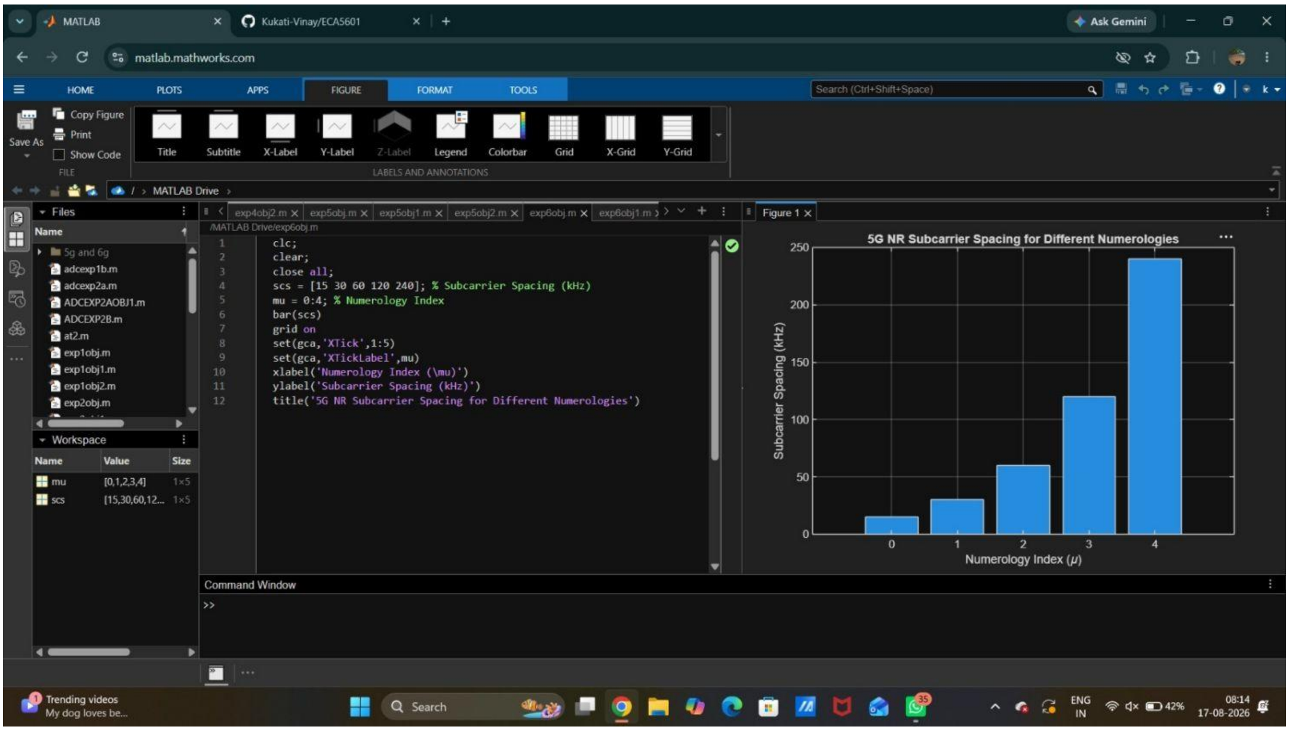
set(gca,'XTickLabel',mu)

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

title('5G NR Subcarrier Spacing for Different Numerologies')

Output:



Objective2:

Procedure

- Define the slot duration corresponding to each subcarrier spacing.
- Plot slot duration versus subcarrier spacing.
- Observe the inverse relationship.

Program:

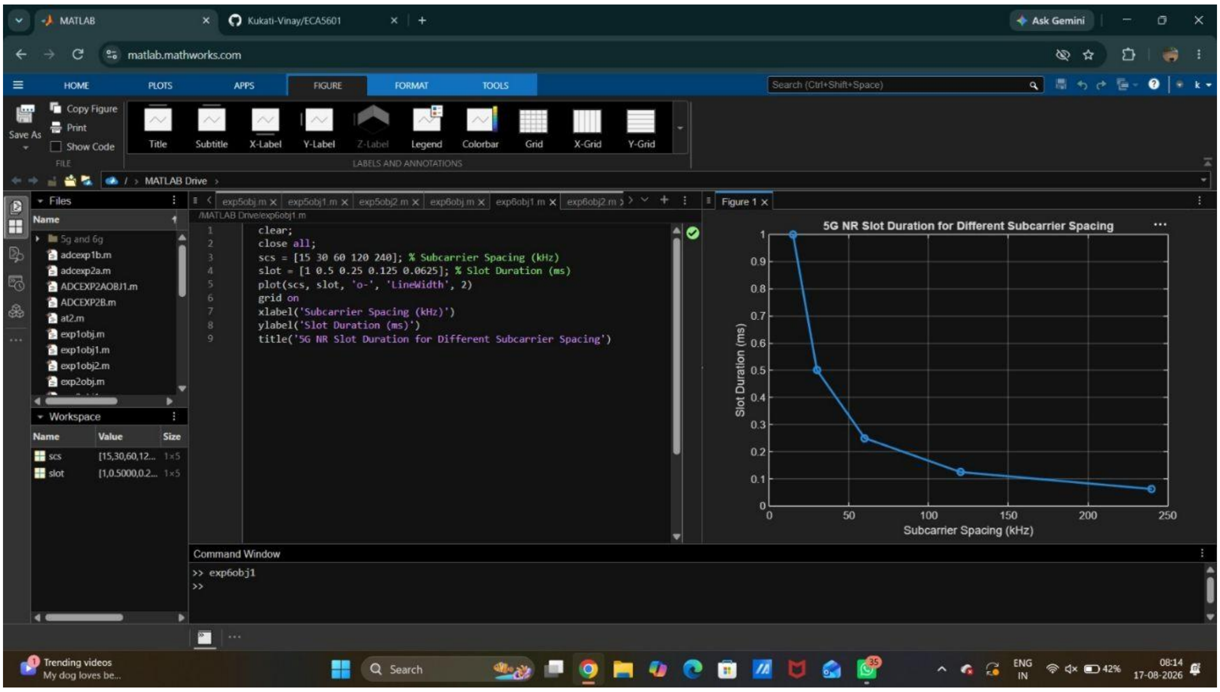
clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

```
slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)
plot(scs, slot, 'o-', 'LineWidth', 2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
```

Output:



Objective3:

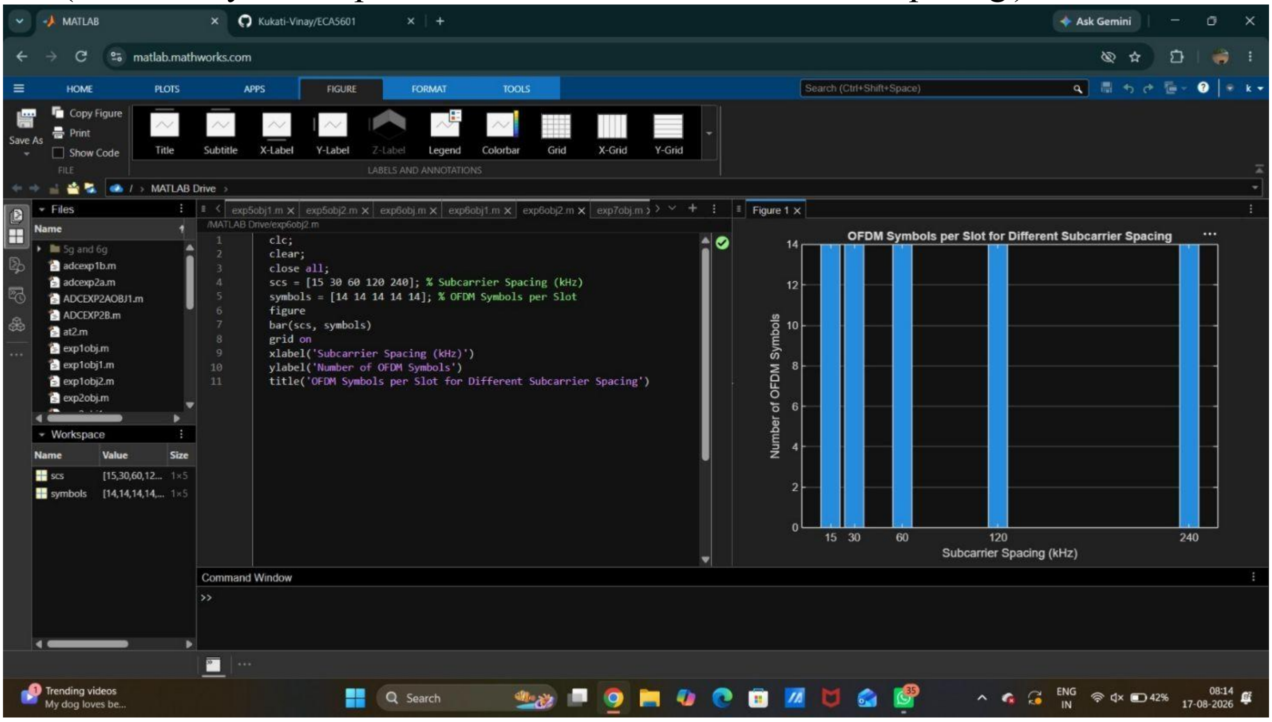
Procedure

- Define the number of OFDM symbols in a slot.
- Plot the OFDM symbols for different numerologies.
- Observe that the number of symbols remains constant.

Program:

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
```

```
symbols = [14 14 14 14 14]; % OFDM Symbols per Slot
figure
bar(scs, symbols)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
```



Result

The MATLAB analysis successfully demonstrated the flexible numerology of 5G NR. The results showed that increasing the subcarrier spacing decreases the slot duration while maintaining a constant number of OFDM symbols per slot under the normal cyclic prefix configuration.